INCH-POUND

MIL-DTL-27190C
19 December 2006
SUPERSEDING
MIL-I-27190B
31 October 1967

DETAIL SPECIFICATION

INDICATOR, PRESSURE AGU-3/A, HYDRAULIC, 0-4,000 PSI

Inactive for new design after 24 March 1999.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hermetically sealed, remote-indicating, synchro-type, integrally lighted, 0-4,000 pounds per square inch (psi) hydraulic pressure indicator, designated AGU-3/A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-595	- Colors Used in Government Procurement (Color chips 37038 and 37875).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MS3134	- Connector, Receptacle, Electric, Solder Type, Jam Nut Mounting, Push-Pull Coupling, Series 1 (Class E, P, J and H).
MS3137	- Connector, Plug, Electric, Solder Type, Push-Pull Coupling Series I (Class E, P and J).
MIL-DTL-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-S-7742	- Screw Threads, Standard, Optimum Selected Series: General Specification for.
MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys.
MIL-T-25307	- Transmitter, Pressure, Hydraulic, Type MJ-1.
MIL-L-27160	- Lighting, Instrument, Integral, White General Specification for.
MS33639	- Cases, Instrument, Clamp-Mounted, Aircraft.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-889	- Dissimilar Metals.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-781	- Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification and Production.
MIL-HDBK-831	- Preparation of Test Reports.

(Copies of these documents are available at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.2.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

STANDARDIZATION DOCUMENTS

SD-6 - Provisions Governing Qualification.

(Copies of these documents are available at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ASME INTERNATIONAL

ASME Y14.100 - Engineering Drawing Practices.

(Copies of these documents are available at http://www.asme.org/ or from ASME International, Three Park Avenue, New York, NY 10016-5990.)

SAE INTERNATIONAL

SAE AS20708 - Synchros, General Specification for.

(Copies of these documents are available at http://www.sae.org/ or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Hydraulic pressure indicators furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.2 Materials.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

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3.2.2 Non-magnetic materials. Non-magnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.2.3 Non-ferrous materials. Non-ferrous materials shall be used for all parts of the indicator, except where ferrous materials are essential.

3.2.4 Metals. Metals shall be corrosion-resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.2.5 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.6 Toxic and corrosive fumes. The indicator shall be of such material and construction that when burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be released.

3.2.7 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3 Design and construction. The indicator shall be designed for use with the type MJ-1 transformer conforming to MIL-T-25307.

3.3.1 Case. The case shall be hermetically sealed and the dimensions shall conform to MS33639 for the 1-inch size. Dimension L shall be 3.00 inches ± 0.25 inch.

3.3.1.1 Cap. A separate nonhermetically sealed cap may be used on the back of the case for installing the electrical receptacle. The cap shall be made of nonferrous low-density metal.

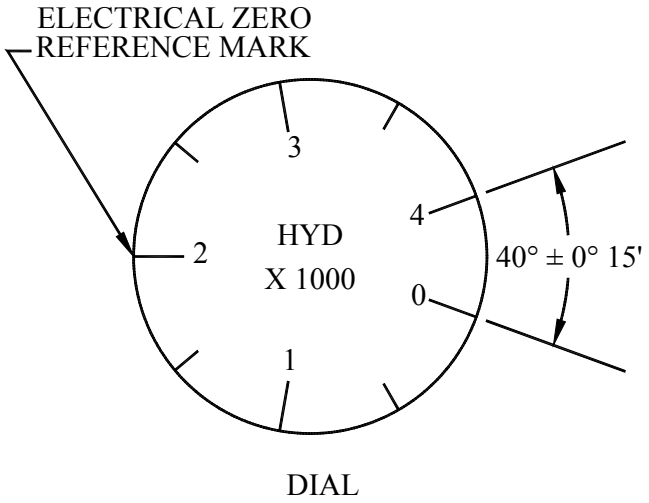
3.3.1.2 Case-to-cover glass distance. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

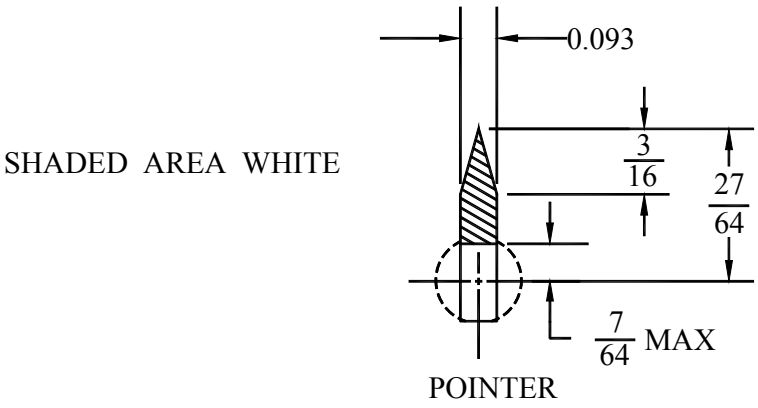
3.3.1.3 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain no more then 0.006 milligram (mg) of water vapor per liter (dew point -65 °C) at the filling pressure. The filling medium shall be 88 percent to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 psi to 14.7 psi. Other concentrations of helium shall not be used except with prior approval of the procuring activity.

3.3.2 Lighting. The dial and pointer shall be integrally lighted in accordance with MIL-L-27160 and shall meet all requirements specified herein.

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3.3.3 Pointer. The pointer shall conform to figure 1 and shall be suitable for integral lighting. The length shall be such that the tip of the pointer will extend into the scale a distance equal to 1/3 to 2/3 the length of the shortest graduation.





Dimensions in inches.
Unless otherwise specified, tolerances: fractions ±1/64; decimals ±0.010.

FIGURE 1. Dial and pointer.

3.3.3.1 Pointer finish. The lighted portion of the pointer shall be finished in accordance with FED-STD-595, color number 37875.

3.3.4 Dial. The dial shall conform to figure 1. A minimum diameter of 15/16 inch shall be maintained across the outside of the graduations. The 2,000-psi graduation shall be located on the horizontal centerline at the left side of the dial. The graduation for each 500-psi increment shall be 40 degrees ± 0 degrees 15 minutes apart.

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3.3.4.1 Dial markings. The dimensions and colors of the dial markings shall be in accordance with table I. Colors shall conform to FED-STD-595.

TABLE I. Dial marking dimensions and colors.

	Height or length in inches ±1/64 inch	Width of line or graduation in inches ±0.005 inch	Color number
Numerals 0, 1, 2, 3, 4	3/32	0.0213	37875
Graduations at 0, 1,000, 3,000, 4,000	7/64	0.030	37875
Graduation at 2,000	3/32	0.030	37875
500-psi graduations	1/16	0.015	37875
Lettering "HYD"	5/64	0.015	37875
Lettering "X1000"	5/64	0.010	37875
Background of dial	--	--	37038

3.3.4.2 Finish of markings not specified. The finish of markings not otherwise specified shall be in accordance with FED-STD-595, color number 37038.

3.3.4.3 Dial visibility. Portions of the minor graduations on the dial shall be visible when viewed from any position within the frustum of a cone, the side of which intersects the outer edge of the cover glass and makes an angle of 30 degrees with the perpendicular through the center of the dial.

3.3.4.4 Dial-to-cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.188 inch.

3.3.5 Filler tube. A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface. If a tube is used, it shall be of such length as to permit three refillings of the case.

3.3.6 Cover glass. The cover glass shall be clear, flat, and free from flaws that would interfere with the normal reading of the instrument. The thickness of the cover glass shall be the minimum practicable and suitable for the purpose.

3.3.7 Fasteners. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be located so that they will not interfere with dial markings or prevent the markings from being located in the proper position.

3.3.8 Electrical zero reference mark. The dial shall be provided with an electrical zero reference mark on the scale portion equidistant in angular degrees between the first and last dial graduations.

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3.3.9 Electrical connector. The indicator shall be provided with an electrical connector, MS3134H-7P or equivalent, to mate with MS3137E7-50S or equivalent.

3.3.10 Electrical wiring. The internal and external wiring of the indicator shall be in accordance with figure 2.

3.3.11 Indicator synchro. The indicator synchro shall consist of a salient two-pole wound rotor used as the primary and shall be designated for 26 volt (V), 400-cycle, single-phase power, with Y-connected stators as the secondaries. The stators shall be wound so the amplitudes of the induced voltages measured between any two stator leads will vary approximately sinusoidally with the angular positions of the rotor. Pivots, bearings, and gears shall not bind or shake and shall be as nearly frictionless as possible.

3.3.11.1 Synchro receiver. The synchro receiver used shall have electrical characteristics compatible with those defined in SAE AS20708.

3.3.12 Secondary voltage. The maximum open-circuit voltage measured between any two secondary leads shall be 11.8 V ± 3 percent.

3.3.13 Reliability. The indicator shall have a minimum mean-time-between-failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and a confidence factor of 90 percent when tested as specified in 4.5.24.

3.3.14 Longevity. The indicator shall have an operating life span of 3,000 hours.

3.4 Performance. The indicator shall operate satisfactorily after subjection to the following conditions:

a. Temperatures ranging from -65 °C to 71 °C.
b. Vibration of 2 g from 5 cycles per seconds (cps) to 500 cps.
c. Pressure altitude of 100,000 feet at -55 °C.
d. Eight cycles of alternate immersion in tap water at 85 °C ± 2 °C and 5 °C ± 2 °C.
e. Exposure to salt sea atmosphere for 48 hours without evidence of corrosion or deterioration.

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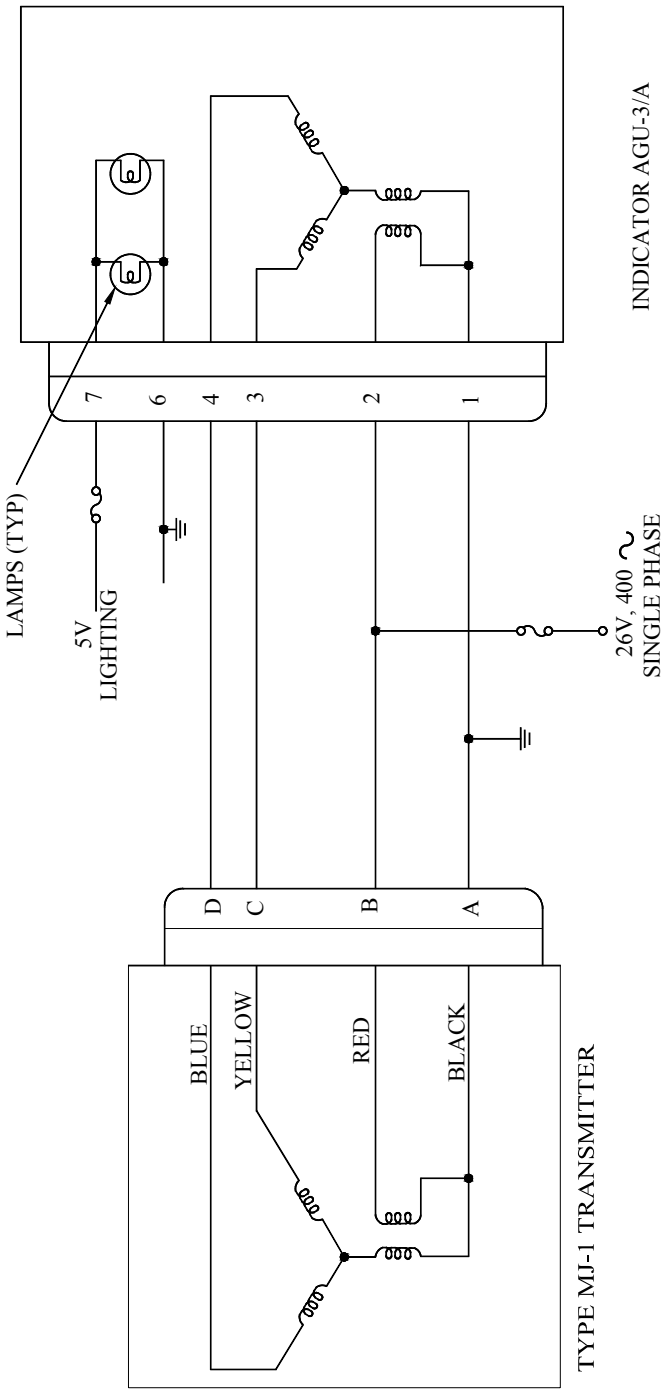


FIGURE 2. Wiring diagram.

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3.4.1 Tolerances. The indicator shall operate within the tolerances specified in tables II and III.

TABLE II. Scale error tolerances.

PSI	Tolerance ±psi
0	50
500	50
1,000	50
1,500	50
2,000	50
2,500	50
3,000	50
3,500	50
4,000	50

TABLE III. Test tolerances.

Test	Tolerance ±psi
Position error at 2,500 psi	25
Friction:	
Room temperature	50
High temperature	50
Low temperature	75
Vibration:	
Pointer oscillation	50
Pointer variation	50

3.4.2 Dielectric strength. The indicator shall withstand 200V root square mean (rms) at a commercial frequency for a period of 5 seconds.

3.4.3 Case leakage. Case leakage shall not exceed 0.1 micron cubic foot per hour.

3.4.4 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is subjected to a temperature of 85 °C ± 5 °C for a minimum period of 2 hours and then the face of the cover glass is immediately immersed in an ice bath for 3 minutes.

3.4.5 Step input response. The time required for the indicator to properly respond to a step input equivalent to a 2,000-psi pressure change shall not exceed 2.0 seconds with not more than two overshoots.

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3.4.6 Voltage and frequency variation. The indicator shall withstand voltage and frequency variations from 22 V to 30 V and 320 cps to 480 cps without exceeding the scale error and friction tolerances specified in tables II and III.

3.4.7 Effect of lead length. When the lead length between the indicator and transmitter is changed from 5 feet to 500 feet, the indication shall not change more than 1 degree of arc. The ground lead shall be kept short.

3.4.8 Magnetic effect. The compass deflection shall not exceed 3 degrees of arc when the indicator is positioned 5-1/2 inches from the compass and the horizontal magnetic field intensity at the indicator is 0.17 oersted to 0.19 oersted.

3.4.9 Unit torque gradient. The unit torque gradient shall be a minimum of 2,000 milligram-millimeters per angular degree of rotation.

3.4.10 Radio noise interference. The indicator shall meet the radio noise interference requirements of MIL-STD-461.

3.5 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of ASME Y14.100 shall govern the manufacturer's part numbers and changes thereto.

3.6 Finishes and protective coatings.

3.6.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except that small holes, straight threads, case inserts, and parts in hermetically sealed enclosures need not be anodized.

3.6.1.1 Resistance to anodizing. Aluminum alloy parts that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-DTL-5541.

3.6.1.2 Abrasion resistance. When abrasion resistance is a factor, chemical films in accordance with MIL-DTL-5541 shall not be used in lieu of anodizing.

3.7 Weight. The weight of the indicator shall not exceed 0.50 pound.

3.8 Screw threads. Unless otherwise specified (see 6.2), the threads of all machine screws shall conform to MIL-S-7742.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.9.1 Nameplate. A nameplate shall be permanently attached to the rear of the case. Where applicable, the information marked on the nameplate shall conform to MIL-STD-130. The nameplate shall not be glued or cemented to the case nor shall metal decals be used except with prior approval by the procuring activity.

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3.10 Workmanship. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.2.1 through 4.2.5.

4.2.1 Standard atmospheric conditions. Wherever the pressure and temperature existing at the time of the test are not specified, that test shall be made at atmospheric pressure (approximately 29.92 inches of mercury (Hg)) and at room temperature (approximately 25 °C). When tests are made with atmospheric pressure or room temperature differing from these values, proper allowance shall be made for the difference from the specified conditions.

4.2.2 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position with the plane of the dial vertical.

4.2.3 Electrical power. Unless otherwise specified, the indicator shall be tested with 26 V ± 1 V, single phase, 400 cps ± 5 cps, AC power, and with 4.5 V ± 0.5 V AC or DC applied to the lighting circuit.

4.2.4 Tapping. Unless otherwise specified, the indicator shall be lightly tapped or vibrated before each reading.

4.2.5 Integral lighting. The integral lighting tests shall be conducted in accordance with MIL-L-27160 in the sequence specified.

4.3 Qualification inspection. The qualification inspections shall consist of all the tests specified under 4.5.

4.3.1 Nonstandard parts data approval. When nonstandard electronic parts are used in the construction of the indicator, nonstandard parts data approval shall be a prerequisite to qualification approval.

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4.3.2 Test samples. The test samples shall consist of 10 indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity. Seven of the samples shall be subjected to the reliability test only and the other three shall be subjected to all the tests except reliability.

4.3.3 Test report. The contractor shall prepare and submit a test report. MIL-HDBK-831 may be used for guidance.

4.4 Conformance inspection. Conformance inspections shall consist of:

a. Individual tests (see 4.4.1).
b. Sampling tests (see 4.4.2).

4.4.1 Individual tests. Each indicator shall be subjected to the following tests.

a. Examination of product (see 4.5.1).
b. Individual tests of MIL-L-27160.
c. Dielectric strength (see 4.5.2).
d. Electrical zero (see 4.5.3).
e. Scale error and friction (see 4.5.4).
f. Position error (see 4.5.5).
g. Sealing (see 4.5.6).

4.4.2 Sampling tests.

4.4.2.1 Sampling plan A. One indicator shall be selected at random from each 100 or less produced on the acquisition order and subjected to the following tests.

a. Individual tests (see 4.4.1).
b. Sampling plan A tests of MIL-L-27160.
c. Vibration (see 4.5.7).
d. Low temperature (see 4.5.8).
e. High temperature (see 4.5.9).
f. Fogging (see 4.5.10).

4.4.2.2 Sampling plan B. Unless otherwise specified, three indicators shall be selected at random from the first 15 on the acquisition order and subjected to the following tests as described under 4.5:

a. Sampling plan A tests (see 4.4.2.1).
b. Sampling plan B tests of MIL-L-27160.
c. Step input response (see 4.5.11).
d. Voltage and frequency variation (see 4.5.12).
e. Secondary voltage (see 4.5.13).
f. Magnetic effect (see 4.5.14).
g. Effect of lead length (see 4.5.15).
h. Temperature-altitude (see 4.5.16).
i. Low temperature exposure (see 4.5.17).
j. High temperature exposure (see 4.5.18).

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k. Thermal shock (see 4.5.19).
l. Salt fog (see 4.5.20).
m. Radio noise interference (see 4.5.21).
n. Unit torque gradient (see 4.5.22).

4.5 Test methods.

4.5.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and markings. Prior to assembly, dials, pointers, and synchros shall be inspected as to markings, dimensions, and finishes.

4.5.2 Dielectric strength. A potential of 200 V at a commercial frequency shall be applied between any pin of the electrical receptacle and the case of the indicator for a minimum period of 5 seconds. There shall be no breakdown of the insulation.

4.5.3 Electrical zero. The electrical zero test and the verification of leads shall be accomplished on the completed indicator with the case removed. A voltage of 26 V ± 1 V single phase shall be applied to the rotor of the synchro with leads designated R_1 and R_2. R_2 shall be connected to the ground or low potential side of the power input. With a lead designated as S_2 connected to R_2, the following conditions shall exist at electrical zero:

a. The voltage between the remaining stator terminals designated S_1 and S_3 shall be minimal.
b. The voltage between R_1 and S_3 shall be near the low null and shall reach the low null when the rotor is rotated clockwise approximately 30 degrees from electrical zero.

4.5.4 Scale error and friction. The indicator shall be tested for scale error at ambient temperature at the test points specified in table II. The indicator shall be lightly tapped or vibrated before each reading to remove normal instrument friction. The scale errors shall not exceed the tolerances specified in table II. The friction error test may be conducted in conjunction with the scale error test. The difference between readings before and after the indicator is tapped or vibrated shall constitute friction error. Friction error shall not exceed the tolerances specified in table III. This procedure shall be accomplished both up-scale and down-scale at the test points specified.

4.5.5 Position error. The indicator pointer shall be positioned at the electrical zero reference mark with power applied. The change in pointer position when the indicator is held in any position shall not exceed the value specified in table III.

4.5.6 Sealing. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

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4.5.7 Vibration. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, method 514, procedure I. A synchro signal shall be maintained on the stators of the indicator synchro to provide an indication of 2,000 psi. Pointer oscillation shall not exceed 200 psi double amplitude. Upon completion of this test, the indicator shall be subjected to the scale error test at room temperature. The scale errors shall not exceed the tolerances specified in table II.

4.5.8 Low temperature. The indicator shall be subjected to a temperature of -55 °C ± 2 °C for a period of 4 hours. While at this temperature, it shall be checked for scale error and friction. The tolerances shall not exceed those specified in tables II and III. In every case, the indicator shall not require more than 3 seconds to come within the tolerance specified.

4.5.9 High temperature. The indicator shall be subjected to a temperature of 72 °C for a period of 4 hours. While at this temperature, it shall be checked for scale errors. The scale error and friction shall not exceed those specified in tables II and III. The response time shall not exceed 3 seconds.

4.5.10 Fogging. The indicator shall be placed in a controlled ambient temperature of 85 °C ± 5 °C for a minimum period of 2 hours. Within 10 seconds after removal from the oven, the indicator shall be immersed to a depth of 1/4 inch in an ice water bath maintained at 0 °C to 7 °C. After 3 minutes, the indicator shall be removed from the fluid bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected and any evidence of fogging or damage shall be cause for rejection.

4.5.11 Step input response. The indicator pointer shall be positioned at approximately 0 psi pressure. With excitation power off, the test transmitter shall be positioned to the equivalent of 2,000 psi. Excitation power shall then be applied and the time required for the indicator to indicate 2,000 psi ± 100 psi shall not exceed 2.0 seconds. The pointer shall move smoothly and shall not oscillate and shall not overshoot more than two times.

4.5.12 Voltage and frequency variation. The indicator shall meet the test specified in 4.5.11 and the scale error and friction tolerances specified in tables II and III when the voltage and frequency are varied from 22 V to 30 V and 320 cps to 480 cps or any combination thereof.

4.5.13 Secondary voltage. The maximum open-circuit voltage shall be measured between any two secondary leads and shall be 11.8 V ± 3 percent. The variation between the three maximum secondary voltages on the indicator synchro as measured between any two stator leads shall not exceed 0.2 V.

4.5.14 Magnetic effect. The indicator, first not operating and second, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0 degrees, 45 degrees, 90 degrees, 135 degrees, 180 degrees, 225 degrees, 270 degrees, and 315 degrees from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 oersted to 0.19 oersted. Deflection of the compass at any of the specified positions shall not exceed 3 degrees.

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4.5.15 Effect of lead length. The indicator shall not change more than 1 degree of arc when the lead length between the indicator and the transmitter is changed from 5 feet to 500 feet of number 20 AWG gage wire. Equivalent resistances may be substituted for the added wire length. The ground lead shall be kept short.

4.5.16 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, method 504, procedure I (except the temperature shall be -55 °C) with a chamber pressure equivalent to an altitude of 100,000 feet. The rate of pressure change shall exceed an average of 3 psi per second while under these conditions and at the end of a 1-hour period, the scale error and friction test shall be conducted. The errors shall not exceed the tolerances specified in tables II and III. This test may be conducted concurrently with the low temperature exposure test.

4.5.17 Low temperature exposure. The indicator shall be subjected to a temperature of -65 °C ± 2 °C for a period of 48 hours. While at this temperature, the indicator shall be operated to ensure that the pointer moves over the entire scale range without exhibiting sticky or erratic characteristics. The temperature shall then be raised to -55 °C ± 2 °C and held for a period of 24 hours. While at this temperature, the indicator shall be subjected to the scale error and friction test, except that in every case, the indicator shall not require more than 4 seconds to come within the tolerances of tables II and III. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator is again subjected to the scale error and friction test. The scale error and friction shall not exceed the tolerances specified in tables II and III. There shall be no damage or deterioration that will affect subsequent operation.

4.5.18 High temperature exposure. The indicator shall be subjected to a temperature of 71 °C for a period of 24 hours and then tested at this temperature for scale errors as specified in 4.5.4. The scale error tolerances and friction shall not exceed the tolerances specified in tables II and III. All parts of the indicator shall then be maintained at room temperature for a period of not less than 4 hours and the indicator shall then be subjected to the scale error and friction test. The scale errors observed shall not exceed the tolerances specified in table II. There shall be no damage or deterioration that might affect subsequent operation.

4.5.19 Thermal shock. The indicator shall be subjected to eight cycles of immersion in water at a temperature of 85 °C ± 2 °C and then in water at a temperature of 5 °C ± 2 °C without evidence of moisture penetration or damage to the coating or the case. The length of time for each immersion shall be 30 minutes, with not more than 5 seconds between immersions. After eight continuous cycles have been completed, the indicator shall be subjected to the test specified in 4.5.6.

4.5.20 Salt fog. The indicator shall be subjected to a salt fog test for a period of 48 hours in accordance with MIL-STD-810, method 509, procedure I. There shall be no evidence of external corrosion or deterioration that might affect subsequent operation. The scale errors after this test shall not exceed the tolerances specified for the room temperature scale error.

4.5.21 Radio noise interference. The indicator shall be subjected to a radio noise interference test in accordance with MIL-STD-461 except that the frequency range for radiated noise shall be within 0.15 megacycle to 150 megacycles. During this test, the synchro shall be operated at 30 revolutions per minute (rpm) ± 5 rpm.

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4.5.22 Unit torque gradient. The indicator shall be connected to the master transmitter synchro and the master transmitter locked in the electrical zero position. The synchro rotors shall then be energized with rated voltage and frequency ±1 percent. Weights shall be applied to a torque pulley which is positioned on the rotor shaft to rotate the shaft in a counter clockwise direction through an angle of 5 degrees of arc. The torque tending to return the shaft to its original position shall be computed and shall be no less than 2,000 milligram-millimeters per degree of displacement.

4.5.23 Lighting. Lighting tests shall be conducted in accordance with sampling plan B tests of MIL-L-27160, except the dielectric test shall not be conducted.

4.5.24 Reliability. Seven indicators shall be tested for reliability. Recording, data handling, and reporting shall be performed for the government. MIL-HDBK-781 may be used for guidance.

4.5.24.1 Reliability test. The test equipment and fixtures shall include provisions for vibration and temperature cycling at the periodic intervals specified herein. During this test, input signals shall be applied to the indicator at a rate of 2 to 3 cycles (minimum and maximum) per minute from 0 percent to 85 percent of full scale and return. The test shall begin when rated power is applied. After completion of 4 hours of operation with rated power on, which shall include at least 480 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the indicator shall be vibrated at 2.2g ± 10 percent peak acceleration value at any non-resonant frequency between 20 cps and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The temperature cycles shall be conducted in the following sequence: 4 hours at -20 °C, 14 minutes for temperature change to 50 °C at a minimum rate of 5 °C per minute; 4 hours at 50 °C, 14 minutes for temperature change to -20 °C, and 4 hours at -20 °C. In case of malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. Upon completion of each 250 hours of cycling, the indicator shall be checked for scale error. The scale error shall not exceed 150 percent of the tolerance in table II at room temperature. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report that contains test procedures, details of any malfunctions, and corrective actions taken. MIL-HDBK-781 may be used for guidance.

4.5.24.2 Longevity. At the completion of the reliability test, testing of three of the indicators shall be continued for a total of 3,000 hours each. The test time accumulated during the reliability test shall be applied to this test.

4.6 Maintenance of qualification. At specified intervals determined by the qualifying activity, the manufacturer must be able to demonstrate that the company still has the capabilities and facilities necessary to produce the QPL items in accordance with this specification and the provisions governing qualification specified in SD-6.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency, or within the military service's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The indicators covered by this specification are intended for use in conjunction with the type MJ-1 hydraulic pressure transmitter in indicating remotely the hydraulic pressure in aircraft systems that require integrally lighted instruments.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. Screw threads, if different (see 3.8).
d. Packaging (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products that are at the time of award or contract, qualified for inclusion in qualified products list QPL-27190, whether or not such are products have actually been listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the federal government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil.

6.4 Subject term (key word) listing.

aircraft
lighted instruments
MJ-1
remote indicating
transmitter

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Air Force - 99

Preparing Activity:
DLA - GS1

(Project 6685-0086-000)

Review Activity:
Air Force - 71

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